

Lab 4 Group Report

Activity Diagrams for MediSure Health Insurance Management System

Group ID	2026-Y2-S1-MLB-B5G2-02
Topic Name	Health Insurance Management System
Submission	Lab Sheet 04 – Activity Diagrams
Tool Used	Draw.io

Report Contents

This report includes group details, each member's main use case scenario, and the corresponding activity diagram for that member's major function.

1. Group Details

No.	Group Member	Student ID	Major Function
01	Lankadhikara L.R.M.M.P.	IT25101639	Policy Management
02	Gunasinghe N.M.	IT25101773	Claims Management
03	De Zoysa A.I.	IT25102600	Underwriting & Risk Assessment
04	Ramanayaka U.K.D.	IT25100714	Premium & Payment Management
05	Karunarathna W.M.K.U.	IT25103657	Benefits & Coverage Plan Management
06	Kavisekara K.M.H.N.	IT25103510	Customer Support & Inquiry Management

2. Report Structure

For each group member, the report repeats the Lab 4 required structure: the member's use case scenario from the previous lab followed by the corresponding individual activity diagram.

01. Policy Management

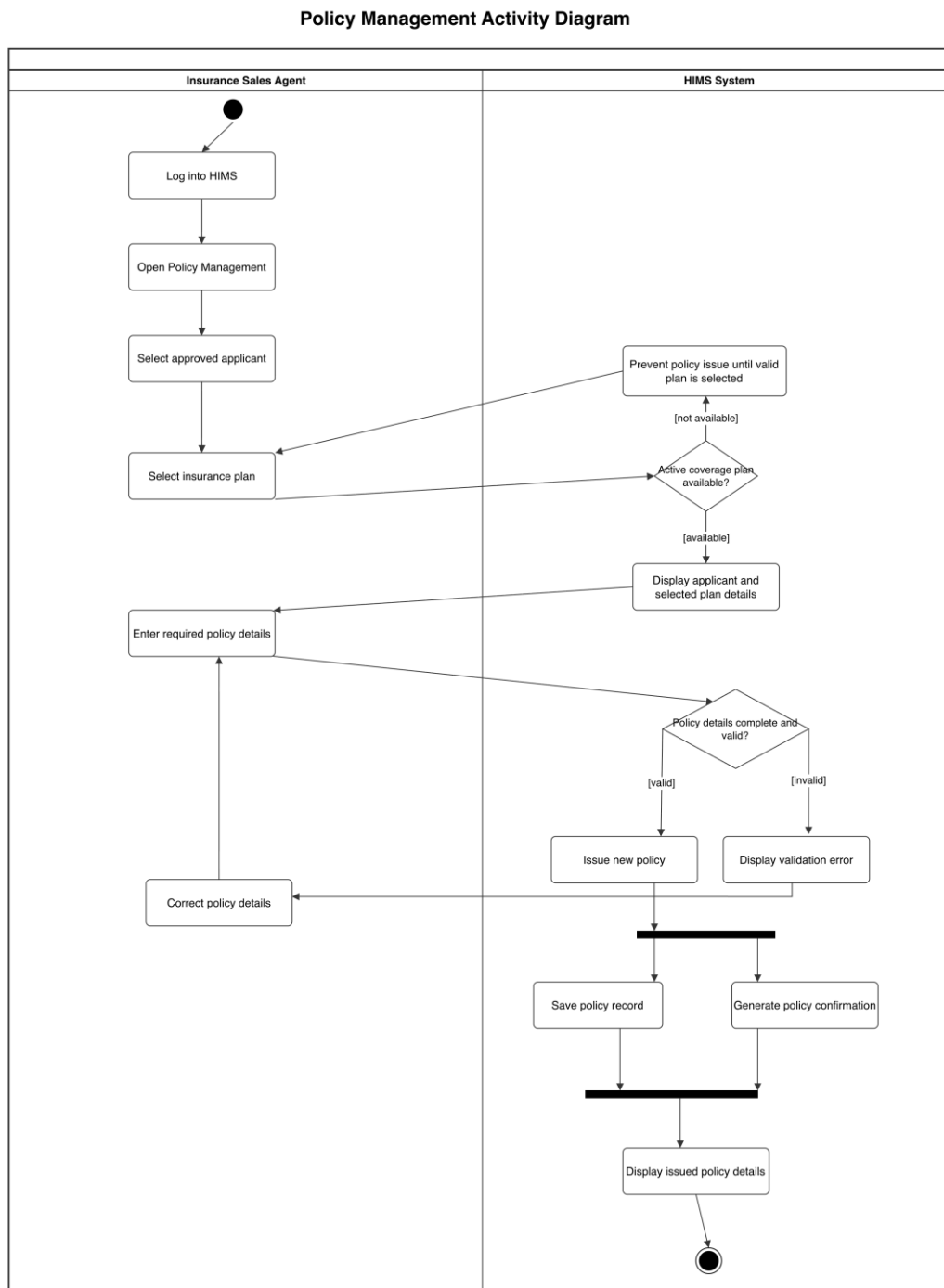
Lankadhikara L.R.M.M.P. - IT25101639

Use Case Scenario

Number	UC-01
Name	Issue New Policy
Summary	An Insurance Sales Agent issues a new health insurance policy for an approved applicant using an active coverage plan.
Priority	5
Preconditions	The Sales Agent is logged into HIMS. The applicant has been approved and at least one active coverage plan exists.
Postconditions	A new policy record is saved and a policy confirmation is generated for the customer.
Primary Actor(s)	Insurance Sales Agent
Secondary Actor(s)	None
Trigger	The Sales Agent selects Policy Management and chooses to issue a new policy.
Main Scenario	
Step	Action
1	The Sales Agent logs into HIMS and opens Policy Management.
2	The Sales Agent selects an approved applicant and selects a suitable active insurance plan.
3	HIMS displays the applicant and selected plan details for confirmation.
4	The Sales Agent enters the required policy details and submits them.
5	HIMS validates the policy details and confirms that all required information is complete.
6	HIMS issues the new policy.
7	HIMS saves the policy record and generates the policy confirmation.
8	HIMS displays the issued policy details to the Sales Agent.
Extensions	
Step	Branching Action
2a	If no active coverage plan is available, HIMS prevents policy issue until a valid plan is selected.
5a	If policy details are invalid or incomplete, HIMS displays a validation error and the Sales Agent corrects the details.
Open Issues	
Issue #	Issue
1	Should newly issued policies require supervisor approval before becoming active?

Corresponding Activity Diagram

Figure 01: Policy Management Activity Diagram



02. Claims Management

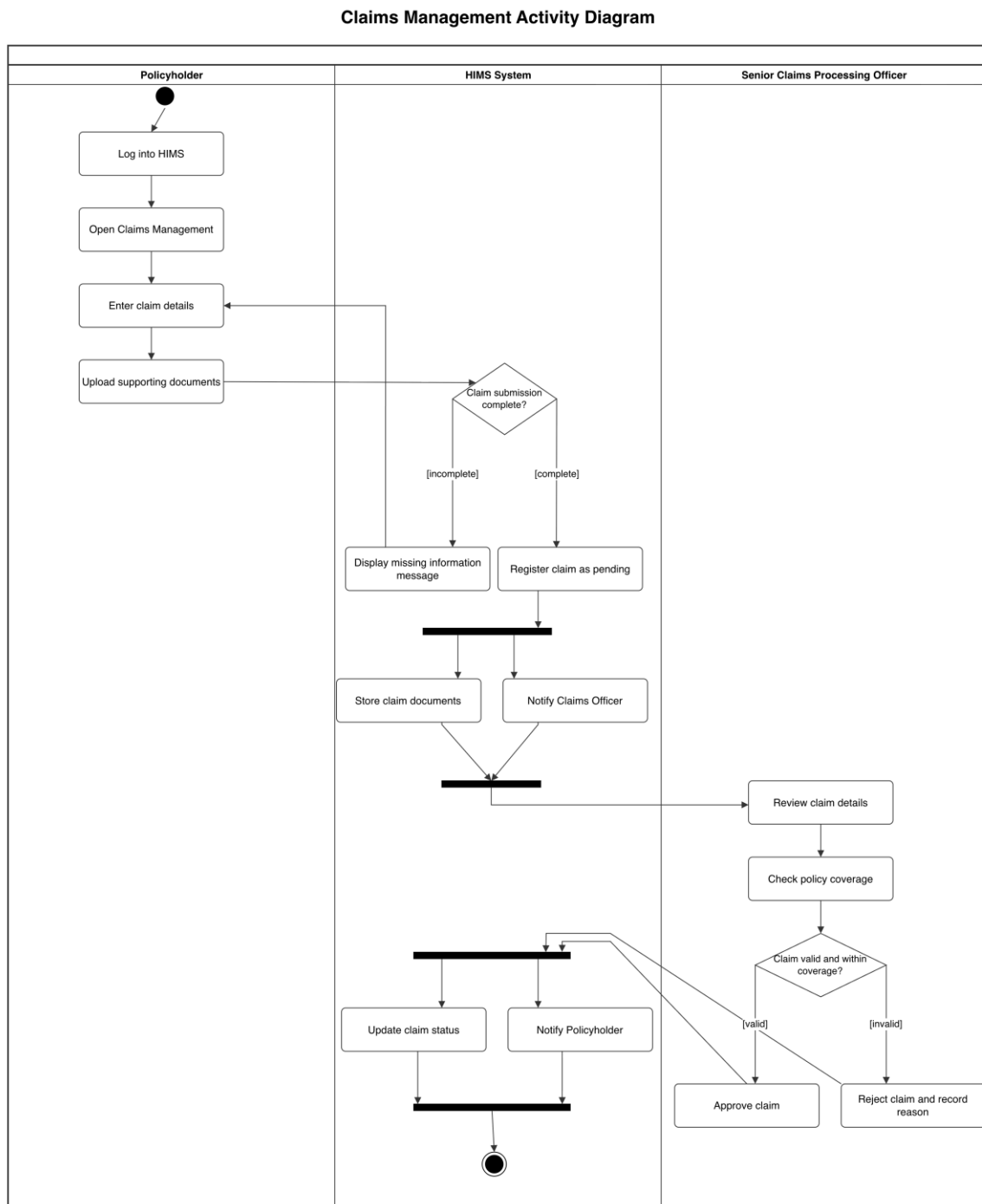
Gunasinghe N.M. - IT25101773

Use Case Scenario

Number	UC-02
Name	Submit and Process Claim
Summary	A Policyholder submits a claim and the Senior Claims Processing Officer reviews it against policy coverage before approving or rejecting it.
Priority	5
Preconditions	The Policyholder has an active policy. The Policyholder and Senior Claims Processing Officer can access HIMS with valid credentials.
Postconditions	The claim is recorded, reviewed, and updated with an approved or rejected status. The Policyholder is notified.
Primary Actor(s)	Policyholder; Senior Claims Processing Officer
Secondary Actor(s)	None
Trigger	The Policyholder opens Claims Management and submits a new claim.
Main Scenario	
Step	Action
1	The Policyholder logs into HIMS and opens Claims Management.
2	The Policyholder enters claim details and uploads supporting documents.
3	HIMS checks whether the claim submission is complete.
4	HIMS registers the claim as pending and stores the claim documents.
5	HIMS notifies the Senior Claims Processing Officer that a claim is ready for review.
6	The Claims Officer reviews the claim details and checks policy coverage.
7	The Claims Officer approves the claim after confirming that it is valid and within policy coverage.
8	HIMS updates the claim status as Approved and notifies the Policyholder.
Extensions	
Step	Branching Action
3a	If the claim submission is incomplete, HIMS displays a missing information message and the Policyholder re-enters the required details.
7a	If the claim is invalid or outside coverage, the Claims Officer rejects the claim and HIMS records the reason.
Open Issues	
Issue #	Issue
1	Should high-value claims require a second approval before final status update?

Corresponding Activity Diagram

Figure 02: Claims Management Activity Diagram



03. Underwriting & Risk Assessment

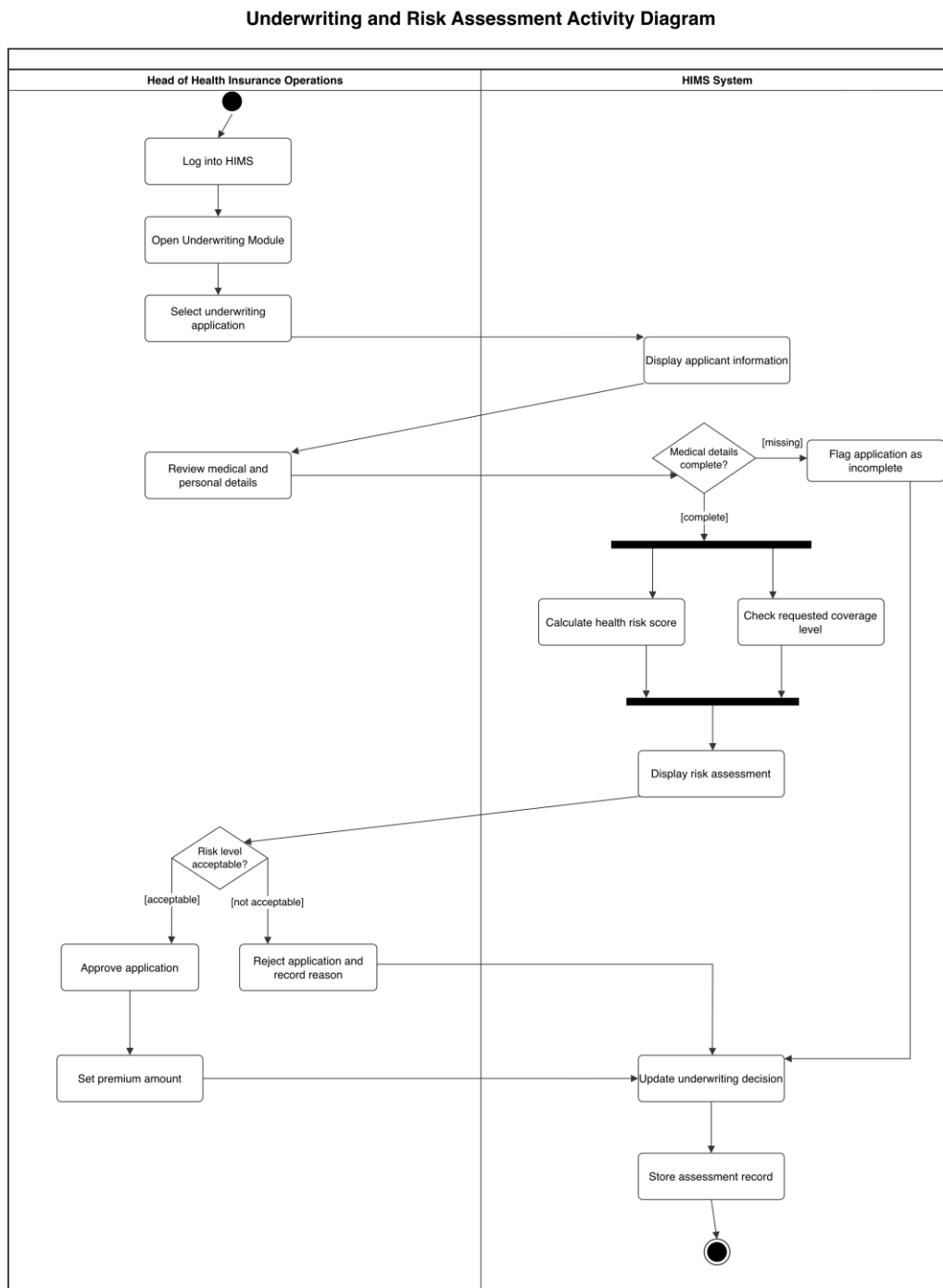
De Zoysa A.I. - IT25102600

Use Case Scenario

Number	UC-03
Name	Assess Underwriting Application
Summary	The Head of Health Insurance Operations reviews an underwriting application, checks risk assessment results, and approves or rejects the application.
Priority	5
Preconditions	The Head of Health Insurance Operations is logged into HIMS, and a pending application with applicant and medical information is available for review.
Postconditions	The underwriting decision and assessment record are stored, enabling or blocking policy issue.
Primary Actor(s)	Head of Health Insurance Operations
Secondary Actor(s)	None
Trigger	The Head of Operations opens the Underwriting Module and selects an application for assessment.
Main Scenario	
Step	Action
1	The Head of Operations logs into HIMS and opens the Underwriting Module.
2	The Head of Operations selects an underwriting application.
3	HIMS displays the applicant information.
4	The Head of Operations reviews medical and personal details.
5	HIMS calculates the health risk score and checks the requested coverage level.
6	HIMS displays the risk assessment.
7	The Head of Operations confirms that the risk level is acceptable.
8	The Head of Operations approves the application and sets the premium amount.
9	HIMS updates and stores the underwriting decision.
Extensions	
Step	Branching Action
4a	If required medical details are missing, HIMS flags the application as incomplete.
7a	If the risk score exceeds the acceptable threshold, the Head of Operations rejects the application and HIMS records the rejection reason.
Open Issues	
Issue #	Issue
1	Should risk score calculation be fully automated or allow manual override?

Corresponding Activity Diagram

Figure 03: Underwriting & Risk Assessment Activity Diagram



04. Premium & Payment Management

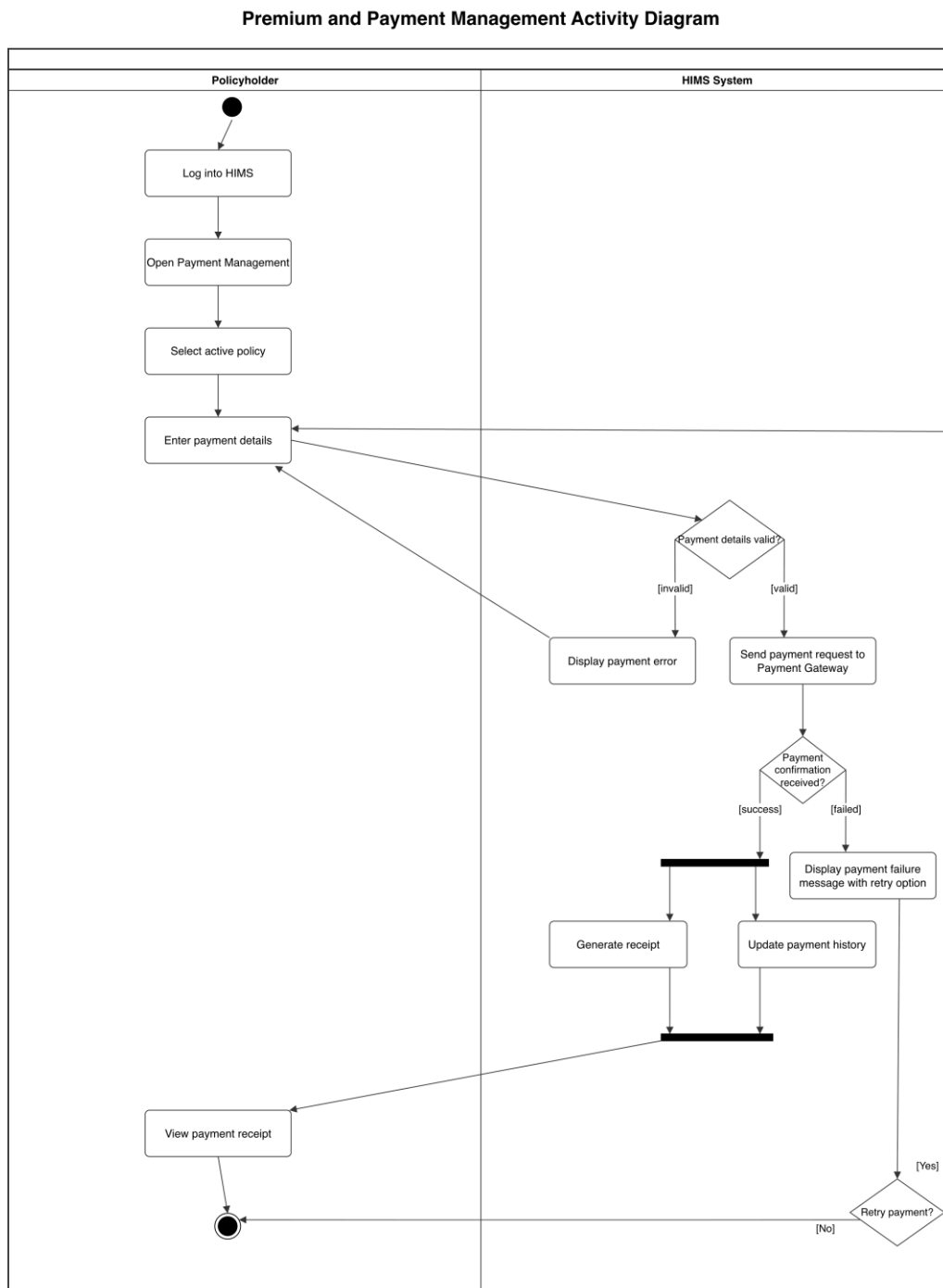
Ramanayaka U.K.D. - IT25100714

Use Case Scenario

Number	UC-04
Name	Make Premium Payment
Summary	A Policyholder pays a due premium through HIMS and receives a payment receipt after successful processing.
Priority	5
Preconditions	The Policyholder is logged into HIMS and has an active policy with a payable premium.
Postconditions	The payment status and payment history are updated, and a receipt is available to the Policyholder.
Primary Actor(s)	Policyholder
Secondary Actor(s)	Payment Gateway
Trigger	The Policyholder opens Payment Management and selects a premium to pay.
Main Scenario	
Step	Action
1	The Policyholder logs into HIMS and opens Payment Management.
2	The Policyholder selects an active policy and enters payment details.
3	HIMS validates the payment details.
4	HIMS sends the payment request to the Payment Gateway for processing.
5	The Payment Gateway confirms that the payment is successful.
6	HIMS generates a receipt and updates the payment history.
7	The Policyholder views the payment receipt.
Extensions	
Step	Branching Action
3a	If payment details are invalid, HIMS displays a payment error and allows the Policyholder to correct the details.
5a	If payment processing fails, the Payment Gateway returns a failure response and HIMS displays a payment failure message with a retry option. If the Policyholder chooses to retry, the flow returns to entering payment details; otherwise, the payment attempt ends.
Open Issues	
Issue #	Issue
1	Should the academic prototype simulate the Payment Gateway response or integrate a sandbox gateway?

Corresponding Activity Diagram

Figure 04: Premium & Payment Management Activity Diagram



05. Benefits & Coverage Plan Management

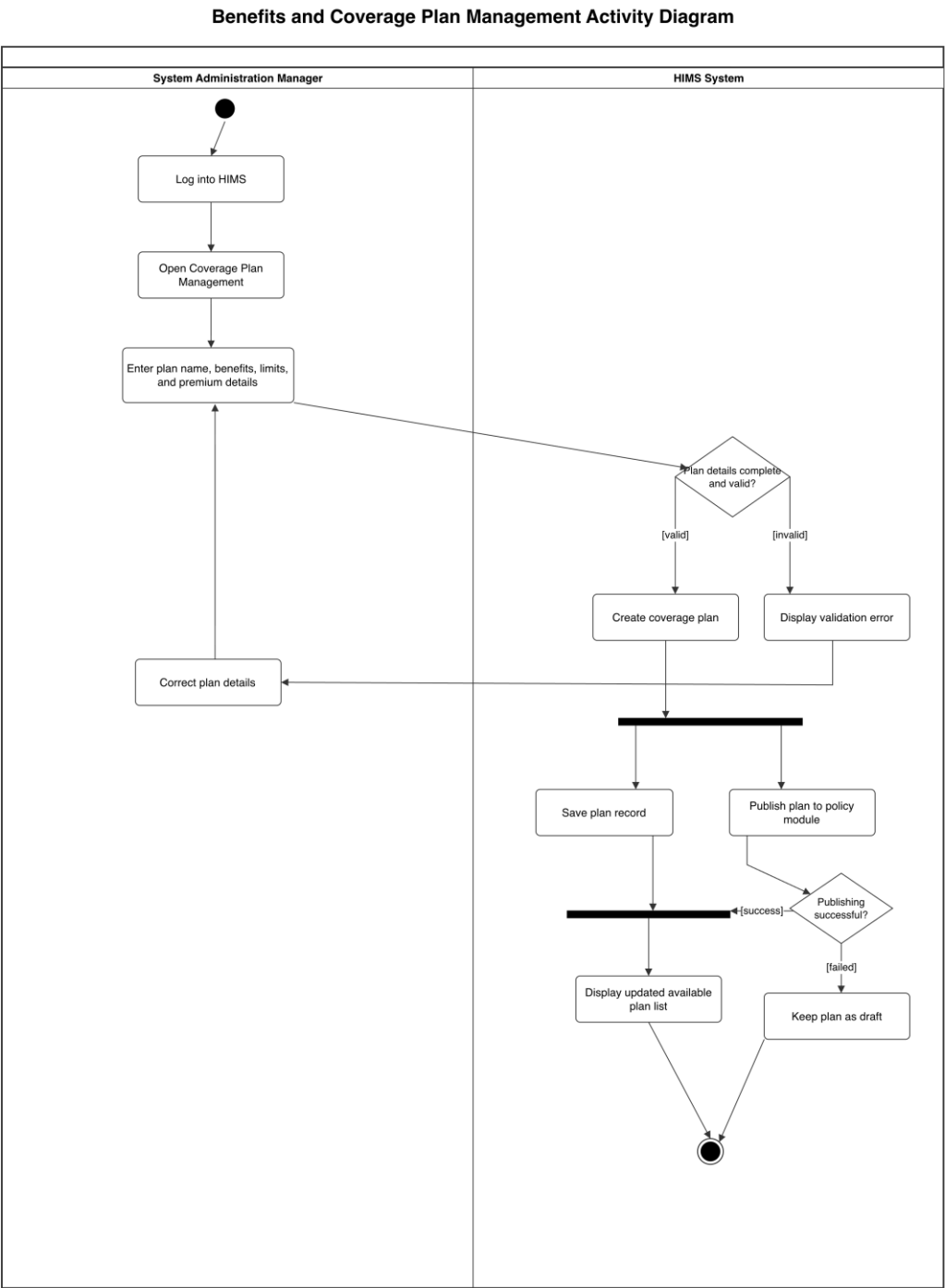
Karunarathna W.M.K.U. - IT25103657

Use Case Scenario

Number	UC-05
Name	Create Coverage Plan
Summary	The System Administration Manager creates a new coverage plan with benefits, limits, and premium details so it can be used for policy issue.
Priority	5
Preconditions	The System Administration Manager has system-level privileges and is logged into HIMS.
Postconditions	The new coverage plan is saved and published to the policy module as an available plan.
Primary Actor(s)	System Administration Manager
Secondary Actor(s)	None
Trigger	The Admin Manager opens Coverage Plan Management and chooses to create a new plan.
Main Scenario	
Step	Action
1	The System Administration Manager logs into HIMS and opens Coverage Plan Management.
2	The Admin Manager enters the plan name, benefits, limits, and premium details.
3	HIMS validates the plan details.
4	HIMS creates the coverage plan after successful validation.
5	HIMS saves the plan record and publishes the plan to the policy module.
6	HIMS displays the updated available plan list.
Extensions	
Step	Branching Action
3a	If mandatory plan fields are missing or invalid, HIMS displays a validation error and the Admin Manager corrects the details.
5a	If publishing fails, HIMS keeps the plan as draft until it can be published successfully.
Open Issues	
Issue #	Issue
1	Should plan changes require approval before becoming visible to sales agents?

Corresponding Activity Diagram

Figure 05: Benefits & Coverage Plan Management Activity Diagram



06. Customer Support & Inquiry Management

Kavisekara K.M.H.N. - IT25103510

Use Case Scenario

Number	UC-06
Name	Handle Customer Support Inquiry
Summary	A Policyholder submits a support inquiry and a Customer Relations Executive reviews, responds to, and resolves the support ticket.
Priority	5
Preconditions	The Policyholder is logged into HIMS, and the Customer Relations Executive is authorized to manage support tickets.
Postconditions	The support ticket status, response, and history are updated, and the Policyholder is notified.
Primary Actor(s)	Policyholder; Customer Relations Executive
Secondary Actor(s)	Relevant Department Officer, if escalation is required
Trigger	The Policyholder submits a support inquiry through HIMS.
Main Scenario	
Step	Action
1	The Policyholder logs into HIMS and opens the support inquiry form.
2	The Policyholder submits inquiry details.
3	HIMS checks whether the inquiry details are complete.
4	HIMS creates a support ticket and assigns it as open.
5	The Customer Relations Executive opens the ticket and reviews the inquiry details.
6	The Customer Relations Executive enters a response.
7	The Executive confirms that the issue has been resolved.
8	The Customer Relations Executive marks the issue as resolved and closes the ticket.
9	HIMS updates ticket status, records ticket history, and notifies the Policyholder.
10	The Policyholder views the support response.
Extensions	
Step	Branching Action
3a	If inquiry details are incomplete, HIMS displays a missing information message and the Policyholder corrects the inquiry.
7a	If the issue cannot be resolved by the Executive, the ticket remains open or is escalated to the relevant officer.
Open Issues	
Issue #	Issue
1	Should policyholders be allowed to reopen a closed support ticket within a fixed time window?

Corresponding Activity Diagram

Figure 06: Customer Support & Inquiry Management Activity Diagram

Customer Support and Inquiry Management Activity Diagram

